# T0 Contact tests



Which joints, part contacts and binder contacts stay reliable? Build on 160–200 gsm card.

H1 BUS A

TP-H1

**Bare hole** · tape only, no eyelet

H2 V+

TP-H2

**Eyelet** · clamps the 5 mm strip

H3 GND

TP-H3

**Eyelet** · on 12 mm tape

H4 BUS B

TP-H4

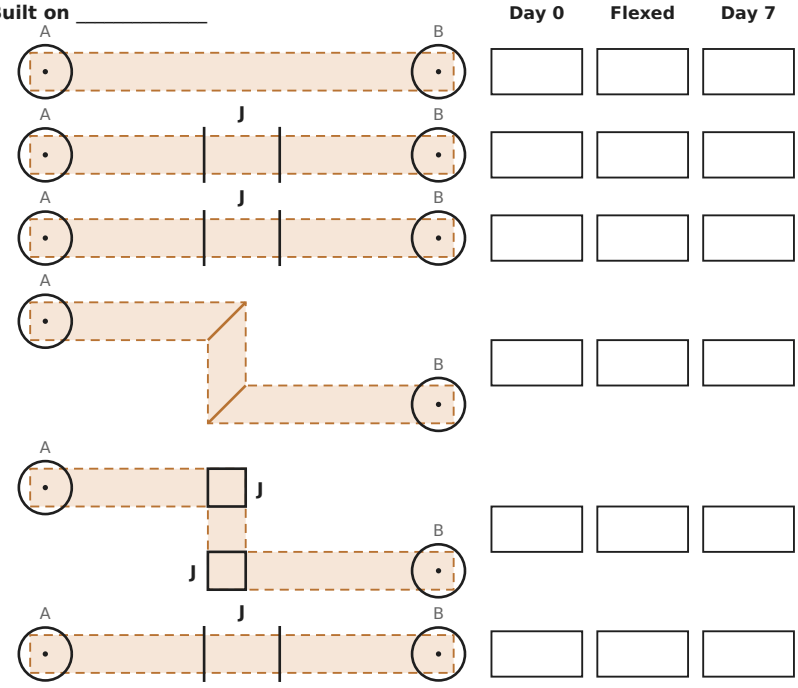
**Eyelet** · soldered to the strip

## 1 Tape joints

Lay each lane from pad A to pad B as marked, then measure A to B in ohms and subtract the meter zero. Write OL for an open circuit. Sheet 2 explains Flexed and Day 7.

**Meter zero** _____ Ω **Built on** _____

| | | Day 0 | Flexed | Day 7 |
|---|---|---|---|---|
| **J1 Continuous** | the reference | | | |
| **J2 Overlap** | cut, overlap 10 mm | | | |
| **J3 Overlap + solder** | solder across it | | | |
| **J4 Folded corners** | fold, don't cut | | | |
| **J5 Cut corners** | cut + overlap at each | | | |
| **J6 Your tape** | e.g. conductive glue | | | |

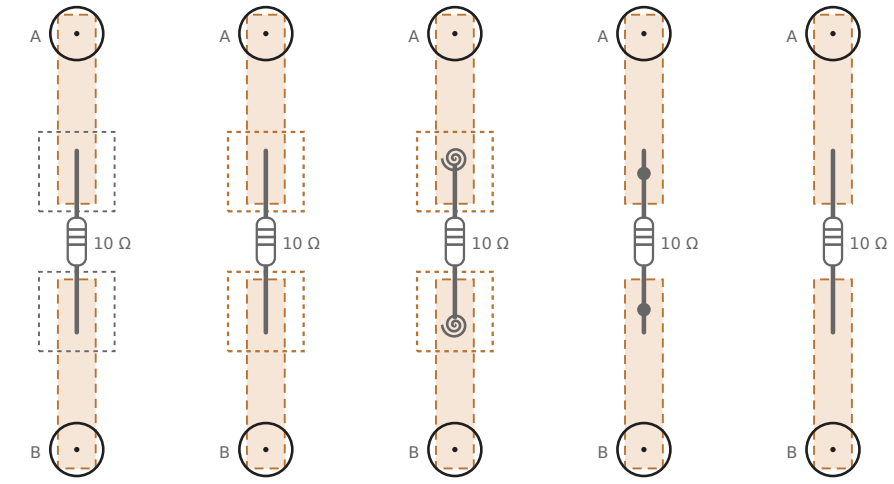



## 2 Part joints

Bridge each gap with a 10 Ω resistor, leads flat on the tape, attached as labelled. Measure A to B. Contact resistance is the reading minus R alone.

**R alone** _____ Ω one resistor, probes on its leads

| | **M1** Flat leads clear tape over | **M2** Flat leads copper patch over | **M3** Spiral leads copper patch over | **M4** Soldered | **M5** Your method |
|---|---|---|---|---|---|
| **Day 0** | | | | | |
| **Flexed** | | | | | |
| **Day 7** | | | | | |
| **Flicker?** | | | | | |




**Notes**

Print at 100 %: this bar should measure 100 mm.

# T0 Running the contact tests



Print this sheet once on plain paper. Print sheet 1 on card three times and sheet 3 once.

H1 BUS A · NC

## You need

- 5 mm copper tape, plus about 15 cm of 12 mm tape for H3
- Five 10 Ω resistors per sheet (brown-black-black)
- Plain brass eyelets for a 6 mm hole, and a setting tool. Not painted aluminium.
- A 2- or 4-hole punch with 80 mm pitch
- A ring binder, and four brass Chicago screws (binding posts) long enough for three sheets
- A multimeter with a 200 Ω range
- A soldering iron for J3, M4 and H4; clear tape; scissors for sheet 3

## Before you start

**1.** Print sheets 1 and 3 at 100 % on 160–200 gsm card. Check a footer bar measures 100 mm with a ruler.

**2.** Lay the four hole lanes (test 3) first, then punch through the crosshairs, through the tape.

**3.** Set eyelets in H2, H3 and H4. Leave H1 bare. Solder H4's eyelet to its strip.

**4.** Zero the meter: 200 Ω range, probes touching. Write the reading on sheet 1 and subtract it from every reading.

H2 V+ · NC

## 1 · Tape joints

**1.** Lay each lane as one piece of 5 mm tape, except at the marks: J means cut and overlap the two pieces by 10 mm.

**2.** J3: solder across the overlap. J4: turn each corner by folding along the printed crease. J5: cut at each corner and overlap.

**3.** Measure from pad A to pad B. Write OL if the meter shows an open circuit.

**4.** **Flexed:** bend the card gently over a table edge 10 times, rub each joint firmly with a thumb 5 times, then measure again.

**5.** **Day 7:** leave the sheet in the binder for a week, then measure again.

## 2 · Part joints

**1.** Measure one resistor on its own and write it as R alone.

**2.** Bridge each station's gap with a resistor, leads flat on the tape. M1: clear tape over flat leads. M2: a copper tape patch over flat leads. M3: coil each lead into a flat spiral, then a copper patch. M4: solder each lead. M5: your own idea.

H3 GND · NC

**3.** Measure A to B. Contact resistance is the reading minus R alone.

**4.** **Flicker:** press on the part while watching the meter. Write Y if the reading jumps.

**5.** Repeat after flexing and on day 7.

10 Ω is low on purpose: a contact of a few ohms shows up clearly against it.

## 3 · Binder contacts

**1.** Build the hole treatments on all three copies, as labelled on sheet 1.

**2.** Bind the three sheets on rings. On each sheet, measure from TP-H*n* to the bare metal of ring *n*. Write the worst of the three readings below.

**3.** **Turning:** keep the probes on and turn the other sheets back and forth 5 times. Write the highest reading, or OL.

**4.** Repeat with Chicago screws, finger-tight.

| | H1 bare | H2 eyelet | H3 12 mm | H4 solder |
|---|---|---|---|---|
| **Rings, still** | | | | |
| **Rings, turning** | | | | |
| **Screws, still** | | | | |
| **Screws, turning** | | | | |

## 4 · Switch contacts

Build, test and record the switches on sheet 3; its steps and pass rule are there.

## Pass or fail

- **Pass:** within 1 Ω of the reference (J1, R alone, or 0 Ω for test 3) and no flicker.
- **Fail:** OL, any flicker, or a reading that drifts by day 7.

Why 1 Ω: lesson circuits use resistors of 100 Ω and up, so 1 Ω changes the current by 1 % or less. Flicker matters more than the exact value.

## Results for the standard

| | |
|---|---|
| **Tape overlaps OK?** | |
| **Lead attachment** | |
| **Hole treatment** | |
| **Binding** | |
| **Switch gap** | |
| **Corner switch OK?** | |

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.

# T0 Switch contacts

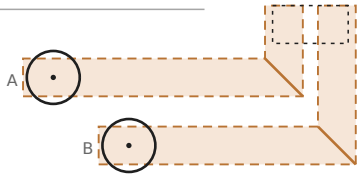


Do fold switches close every time, and does the gap stay open? Build on 160–200 gsm card.

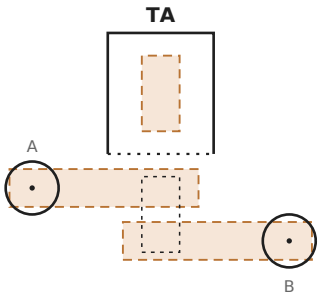
H1 BUS A · NC

## 4 Switch contacts

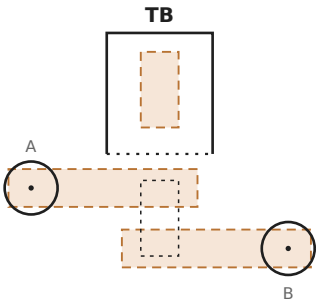
Each switch joins lane A to lane B while its flap is pressed. Measure A to B with the meter on its 200 Ω range.



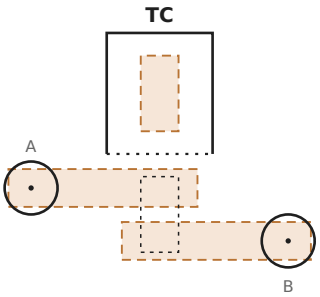

**S1 · corner switch**
fold the corner down to press




2 mm gap, the standard




3 mm gap




your version

H2 V+ · NC

| | Open<br>OL? | Light press<br>Ω | Firm press<br>Ω | Flicker?<br>Y / N | After 20<br>Ω | Day 7<br>Ω |
|---|---|---|---|---|---|---|
| **S1 corner** | | | | | | |
| **TA 2 mm** | | | | | | |
| **TB 3 mm** | | | | | | |
| **TC yours** | | | | | | |

### How to test

1. Lay each switch's A and B lanes as separate pieces of tape. Keep the gap between them clear.
2. S1: stick a 10 mm piece of tape on the corner flap where shown. Tabs: cut the solid lines, crease the dotted hinges and stick a 10 mm piece on each flap.
3. **Open:** with nothing pressed, measure A to B. It must read OL; a number means the gap is bridged.
4. **Light and firm:** rest one finger on the flap and read the meter. Then press hard with a thumb.
5. **Flicker:** keep pressing and rock your finger. Write Y if the reading jumps or drops to OL.
6. **After 20:** press and release 20 times, then take a firm reading. **Day 7:** repeat a week later.

H3 GND · NC

### Pass or fail

- **Pass:** open reads OL, and pressed reads under 10 Ω with no flicker.
- **Fail:** any reading when open, OL or flicker when pressed, or a rise by day 7.

Why 10 Ω: lesson LEDs use 220 Ω, so 10 Ω changes their current by under 5 %. Flicker matters more: it shows up as a flickering LED.

If TB passes as well as TA, standard v0.2 can widen switch gaps to 3 mm and drop the 2 mm exception. Write the result on sheet 2.

**Notes**

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.